

Studying the secondary effects of major wildfires through a case study:

Assessing the PM2.5 Pollution Patterns from the 2017 Thomas fire

Olympe Genty, Timmy Liang, Lauren Scott

Wildfire Impact Strategies (WIS)

5500 Campanile Drive

San Diego, CA 92182

619-594-5200

May 26th, 2025

ABSTRACT

Climate change has significant impacts on the frequency and intensity of natural disasters, including wildfires. In California, the growing frequency and severity of wildfires, exacerbated by prolonged droughts and increased vegetation, pose serious risks to human health and safety. Wildfires not only cause direct harm through property destruction and fatalities but also have secondary health effects, particularly through the dispersion of smoke. Inhalation of wildfire smoke is linked to respiratory diseases and other chronic conditions because of the presence of microscopic particles harmful for the lungs such as the one identified as PM_{2.5}.

This study investigates the impact of major wildfires on PM_{2.5} pollution concentrations in the air through the case study of the December 2017 Thomas Fire. Using air quality data from 2016, a year with minor wildfires, and 2017, the year of the fire, we investigate the impact of the fire on pollution patterns throughout affected California counties (Santa Barbara and Ventura) and unaffected ones (Colusa and Monterey) through statistical methods and singular value decomposition (SVD).

Our results show a marked increase in PM_{2.5} concentrations during the fire in affected regions, especially in December 2017. The spatial analysis from the SVD analysis highlights Santa Barbara and Ventura, the affected counties, as dominant contributors to the PM_{2.5} pattern, underscoring the environmental and public health implications of wildfire-induced air pollution.

INTRODUCTION

Climate change has intensified the frequency and severity of natural disasters, including wildfires. These events disrupt ecosystems, threaten lives, and damage infrastructure. In December 2017, one of the most devastating wildfires in California's history, the Thomas Fire, ignited in Ventura County and quickly spread into neighboring Santa Barbara County. The fire, which raged from December 4, 2017, to January 12, 2018, was fueled by high winds and dry conditions, ultimately burning through 281,893 acres of land. The fire caused extensive property damage, destroying over 1,000 structures and killing two persons. With over 8,500 firefighters mobilized to combat the blaze, the fire's reach extended across multiple communities, leaving behind a trail of destruction and raising significant concerns about public health, particularly from the smoke it generated. [1]

Among the various harmful pollutants released during wildfires, Fine Particulate Matter (PM_{2.5}) is the most concerning for human health. PM_{2.5} includes all particles less than or equal to 2.5 μ m in diameter, small enough to penetrate deep into the lungs and even enter the bloodstream. These particles are primarily produced when vegetation, structures, and other materials are burned in wildfires. The health impacts of PM_{2.5} exposure can be severe and wide-ranging. In the short term, it can cause respiratory issues such as coughing, sore throat, eye irritation, and asthma attacks. Over the long term, repeated exposure to wildfire smoke can lead to chronic conditions such as heart disease, lung damage, reduced lung development in children, and even premature death. [2]

This study focuses on the patterns of PM_{2.5} concentration following the Thomas fire in affected and unaffected California counties, respectfully Santa Barbara and Ventura, and Monterey and Colusa. To draw meaningful conclusions, we compare PM_{2.5} levels in 2017, the year of the

fire, and in 2016, a year that experienced minor wildfire activity across our counties of interest [3]. We chose Monterey and Colusa because neither county was affected by a major wildfire in 2016 or 2017.

Our objective is to identify pollution patterns linked to the Thomas Fire and suggest implications for public health. The methodology employs comparative statistical analysis and SVD to isolate dominant pollution patterns. This research provides a foundation for future studies incorporating health outcomes such as respiratory disease hospitalizations, helping policy makers mitigate fire-related health risks.

METHODS

- Data: PM2.5 concentration

Our data was obtained from the United States Environmental Protection Agency database [4].

We extracted the data corresponding to the pollutant PM2.5 for the counties of Santa Barbara, Ventura, Monterey, and Colusa for the years of 2016 and 2017. Using Python, we created yearly and December-only data matrices for each county for 2016 and 2017 in order to compare their yearly averages and monthly averages.

Then, we created space-time matrices for 2016 and 2017 joining all counties. Some counties did not have consistency in their monitoring sites from one year to another so we selected repeating sites only to have consistency in our results. Our data also presents NaN values. In order to gain accuracy, we chose to select only monitoring locations who had less than 50 NaN values per year. All remaining NaN values are replaced by the yearly average of the location. From this selection process, the remaining locations are:

- Ventura: El Rio-Rio Mesa School, Ojai - East Ojai Ave, Piru - Pacific, Simi Valley-Cochran Street, Thousand Oaks;
- Santa Barbara: Goleta, Santa Maria;
- Colusa: Colusa-Sunrise Blvd, Cortina Indian Rancheria;
- Monterey: Carmel Valley, King City 2, Salinas 3.

We repeated the same procedure from the December data, replacing the NaN values by the December average for the location. The selected locations are:

- Ventura: Ojai - East Ojai Ave, Piru - Pacific, Simi Valley-Cochran Street;
- Santa Barbara: Goleta, Lompoc H Street, Santa Maria;
- Colusa: Colusa-Sunrise Blvd, Cortina Indian Rancheria;

- Monterey: Carmel Valley, King City 2.
- SVD

Singular Value Decomposition (SVD) is a mathematical technique used to break down complex datasets into simpler, interpretable components. In this study, we apply SVD to analyze PM2.5 concentration patterns across selected California counties in 2016 and 2017, with a specific focus on identifying how the Thomas Fire in December 2017 influenced pollution levels.

We construct space-time matrices, where each row corresponds to a monitoring location and each column to a time step (either days of the year or days in December). These matrices are decomposed into three components:

- U : spatial patterns (also known as Empirical Orthogonal Functions or EOFs)
- D : a diagonal matrix of singular values that represent the importance of each mode
- V^T : temporal patterns (also known as Principal Components or PCs)

This decomposition helps isolate the dominant spatial and temporal structures in the PM2.5 data without requiring additional statistical measures.

Matrix U – Spatial Patterns (EOFs)

The matrix U contains spatial patterns across locations. Each column of U , called an Empirical Orthogonal Function (EOF) indicates how strongly a location contributes to a specific mode of variation in the data. The first EOF (EOF1), associated with the largest singular value, represents the most dominant spatial pattern.

In our analysis, EOF1 is used to identify which locations - and by extension, which counties - were most affected by high PM2.5 levels.

Matrix V^T – Temporal Patterns (PCs)

Each row of V^T , called a Principal Component (PC), reflects the temporal evolution of a corresponding spatial mode. The first principal component (PC1) describes the primary temporal pattern in the data.

We plot PC1 for both 2016 and 2017 to examine how PM_{2.5} concentrations vary over time.

However, due to the limited number of days in December, we do not analyze PCs for monthly data; instead, we rely on EOFs to compare spatial differences.

Matrix D – Mode Significance

The diagonal values in D indicate the strength of each mode, guiding us to focus on the most significant patterns (typically the first mode). In this study, we emphasize EOF1 and PC1 to interpret spatial and temporal pollution trends.

RESULTS

- General trends

	2016		2017	
	Yearly	December	Yearly	December
Ventura	8.74	7.16	9.62	83.37
Santa Barbara	6.72	5.88	8.88	16.24
Colusa	6.29	5.30	6.32	6.61
Monterey	5.78	4.41	5.86	6.94

Table 1: Yearly and December PM2.5 concentration average (in $\mu\text{g}/\text{m}^3$) for each studied county in 2016 and 2017. Values superior to $10 \mu\text{g}/\text{m}^3$ are highlighted.

When comparing the yearly average between 2016 and 2017 for each location in **Table 1**, we observe that the affected counties show an increase in their PM2.5 levels compared to the unaffected counties whose averages show insignificant variation - even a small decrease. This observation is exacerbated when looking at the monthly average for December specifically. Ventura county shows a concentration of $7.16 \mu\text{g}/\text{m}^3$ in December 2016, against a concentration of $83.37 \mu\text{g}/\text{m}^3$ in December 2017. Similarly, Santa Barbara shows $5.88 \mu\text{g}/\text{m}^3$ for December 2016 against $16.34 \mu\text{g}/\text{m}^3$ in December 2017. In comparison, Colusa and Monterey respectively show a decrease from 6.61 to $5.3 \mu\text{g}/\text{m}^3$ and from 6.94 to $4.41 \mu\text{g}/\text{m}^3$.

- SVD analysis

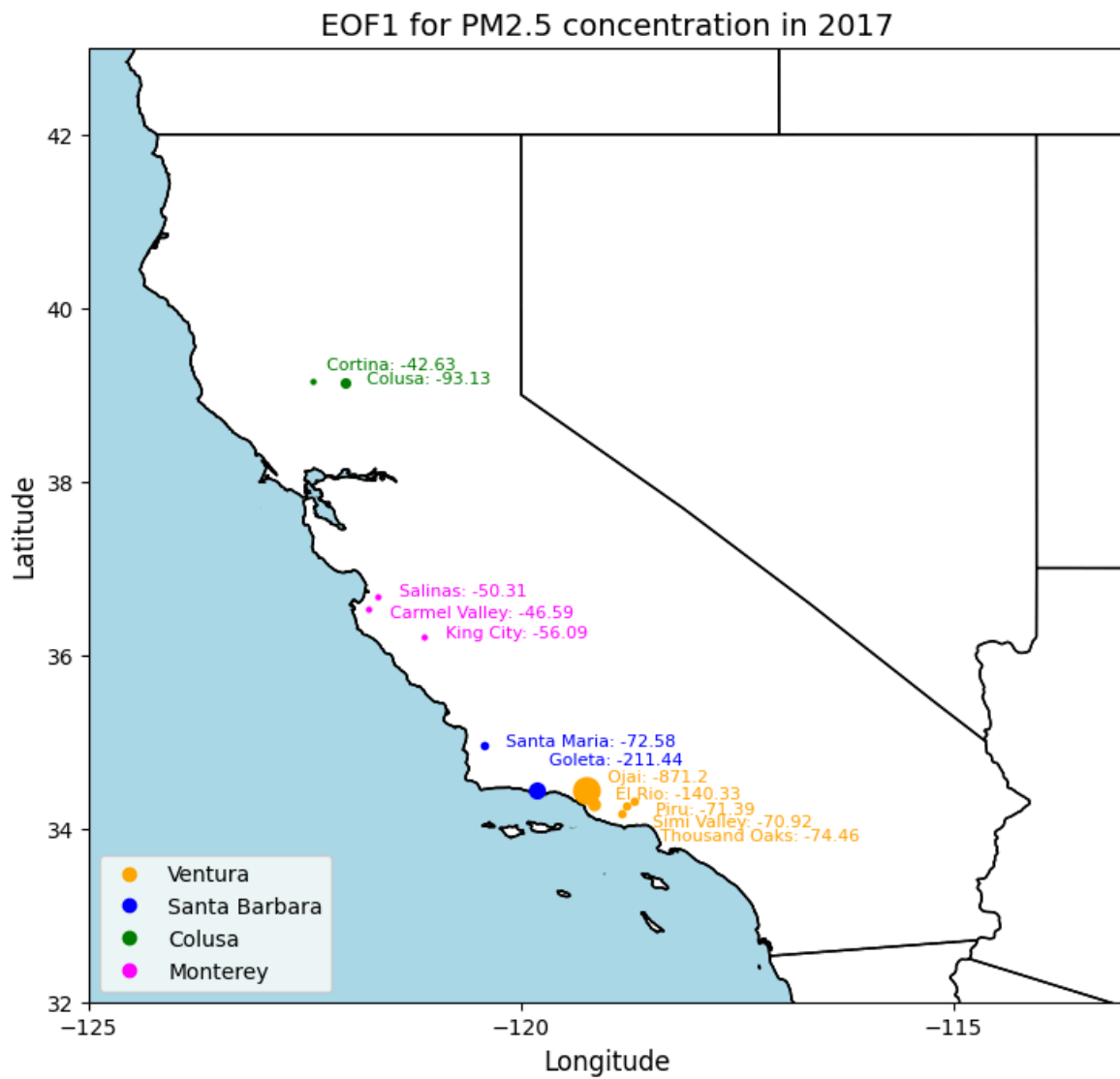


Figure 1: Spatial patterns of the PM2.5 concentration in 2017 over different California counties.

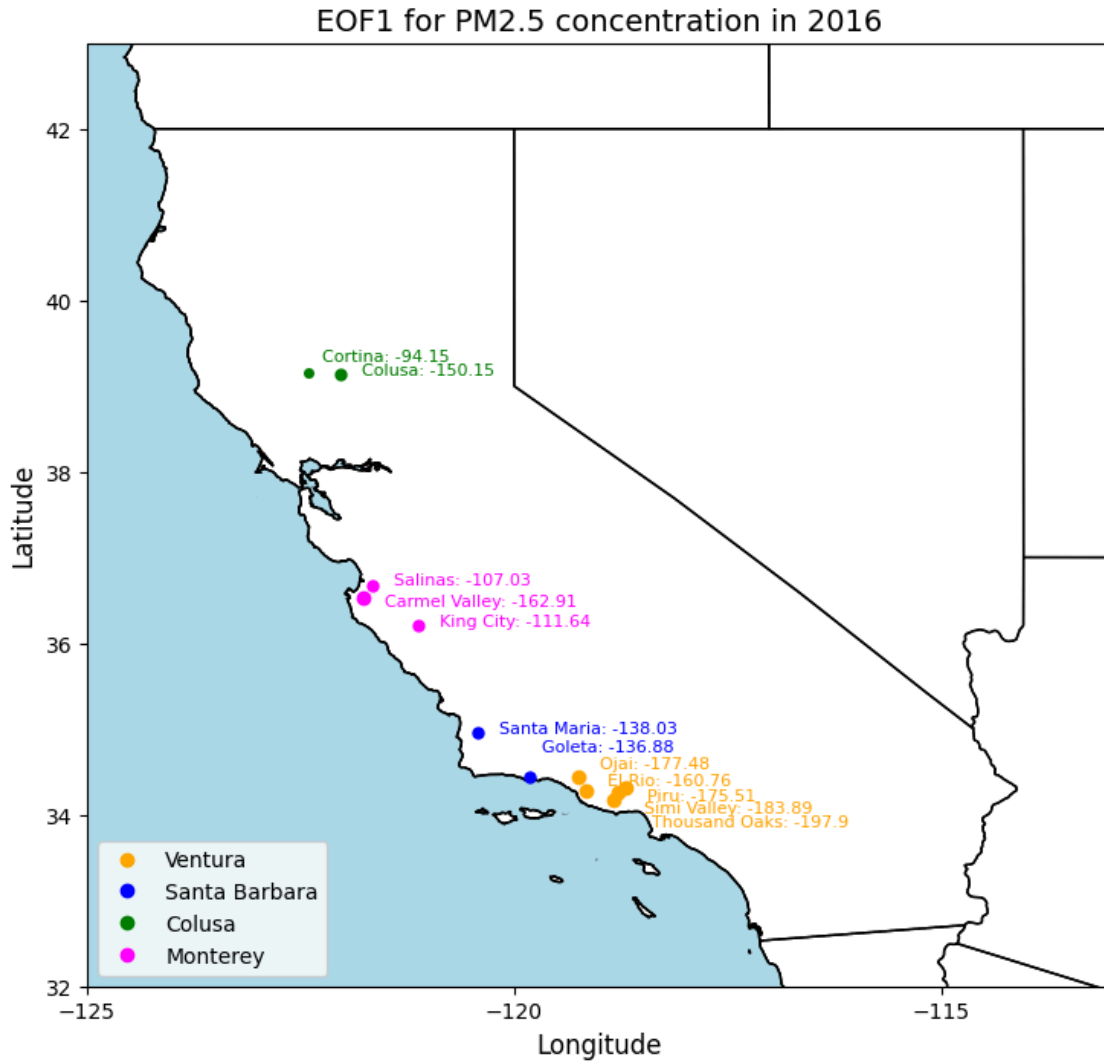


Figure 2: Spatial patterns of the PM2.5 concentration in 2016 over different California counties.

From our yearly EOF1 analysis in **Figure 1**, we observe that in 2017, locations in Ventura and Santa Barbara show the strongest signals in EOF1, with disproportionately high negative values in Ojai and Goleta. This indicates a strong local influence on PM2.5 patterns, which align with areas impacted by the Thomas Fire according to the smoke dispersion in **Figure 3**. In comparison, in **Figure 2**, we observe uniform negative contributions across all counties for 2016.



Figure 3: Satellite image from the Terra satellite of the smoke from the Thomas Fire in Ventura County, on December 5, along with the smoke from 2 other smaller fires, in Southern California.

Image obtained from the NASA Earth Observatory at

<https://earthobservatory.nasa.gov/NaturalHazards/view.php?id=91379>

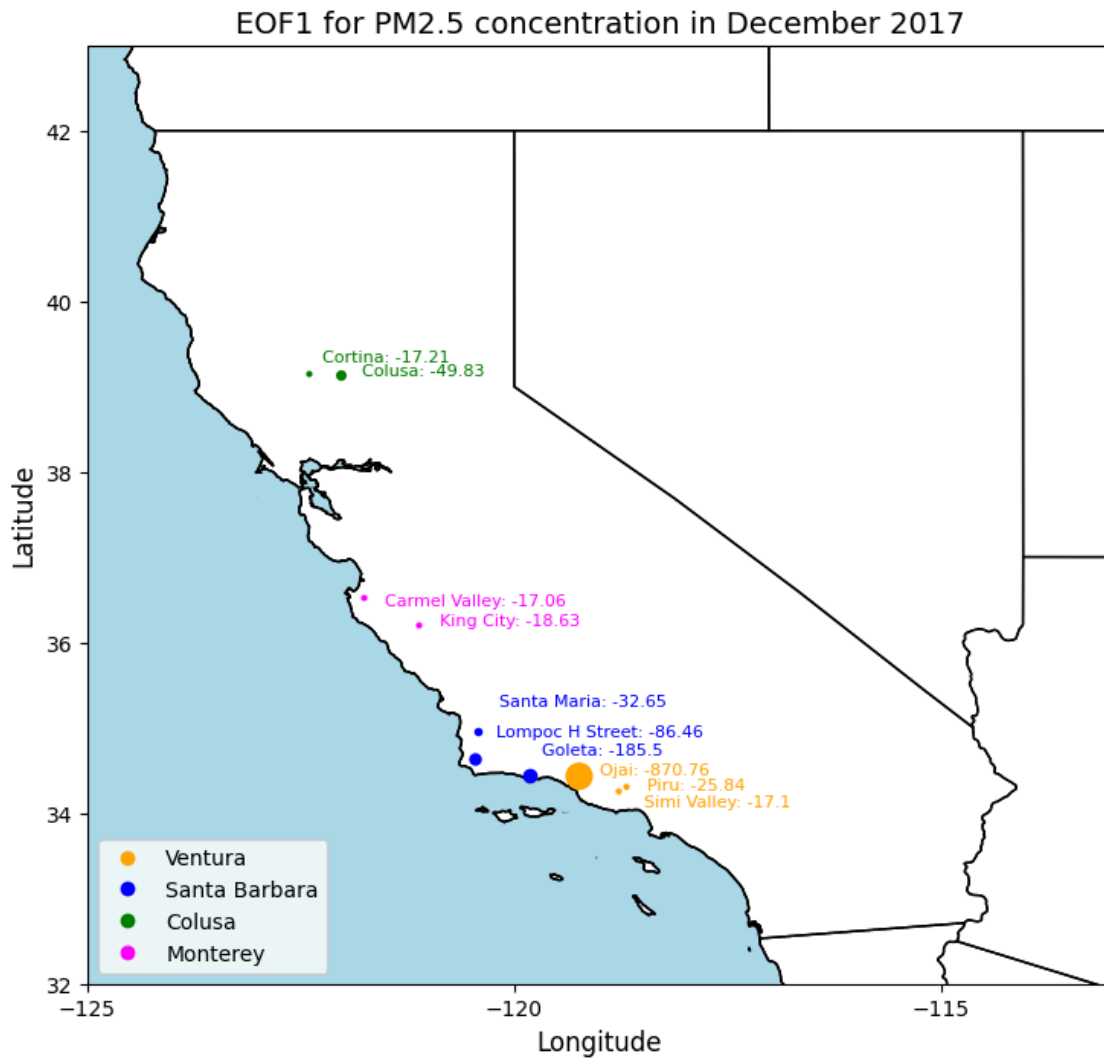


Figure 4: Spatial patterns of the PM2.5 concentration in December 2017 over different California counties.

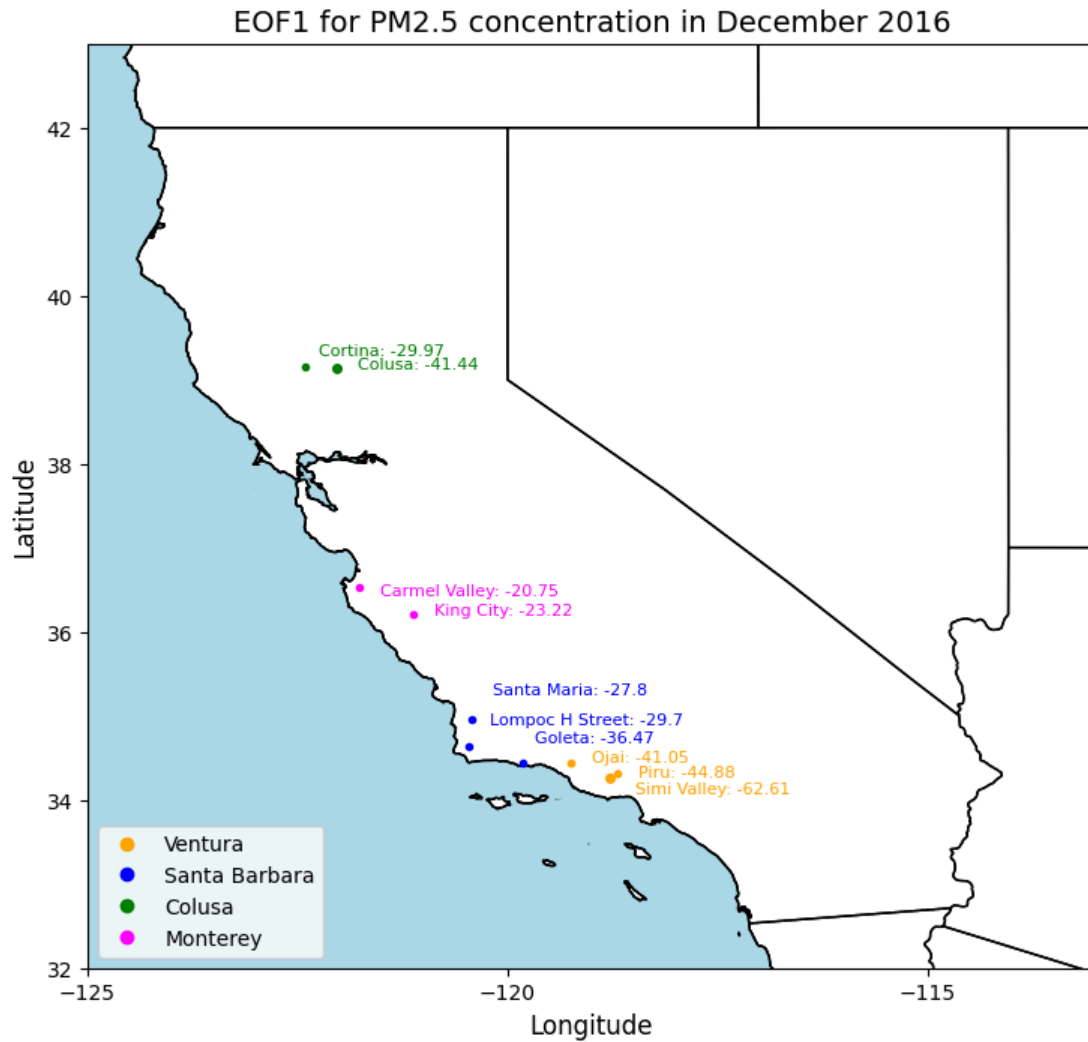


Figure 5: Spatial patterns of the PM2.5 concentration in December 2016 over different California counties.

The December EOF1s for 2016 and 2017 - **Figure 4** and **Figure 5** - mirror our yearly EOF1, reinforcing December's critical role in 2017's pollution patterns.

Our PC1 analysis in **Figure 6** reveals a sharp change in December 2017. This period coincides with the Thomas Fire, highlighting its significant impact. In contrast, in **Figure 7**, 2016 shows relatively stable trends in our PC1 values throughout the year with no distinct anomalies, supporting its use as a baseline.

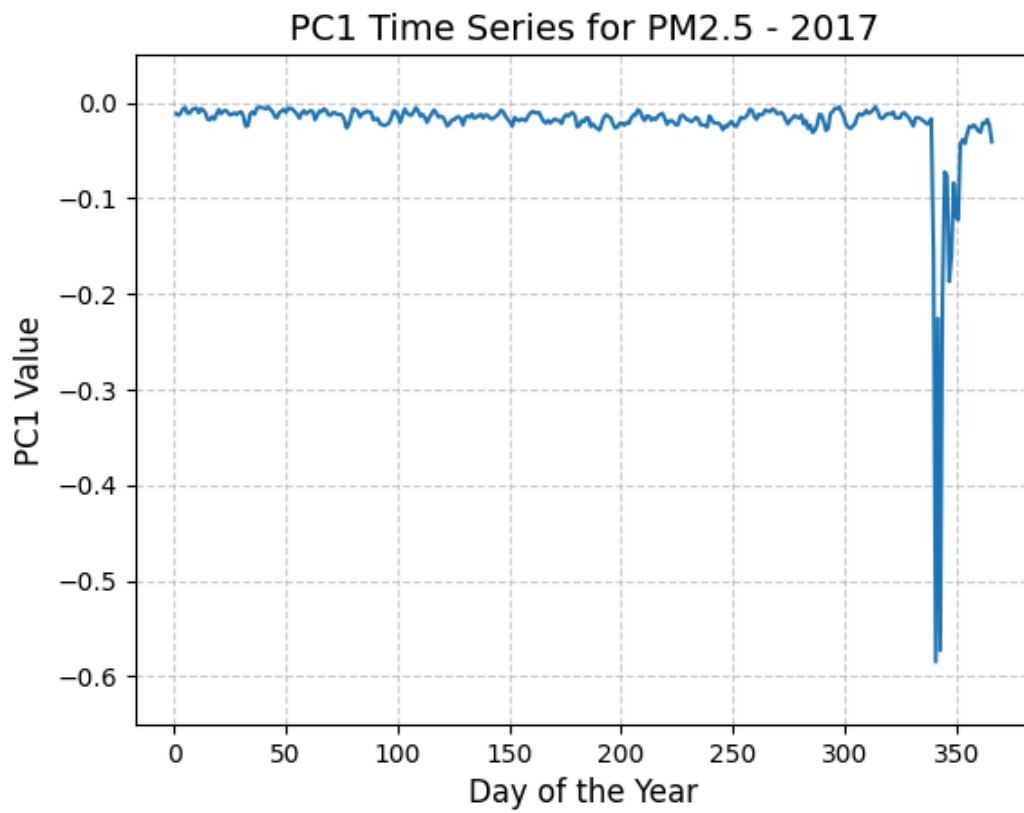


Figure 6: Temporal pattern for PM2.5 concentrations over the year of 2017.

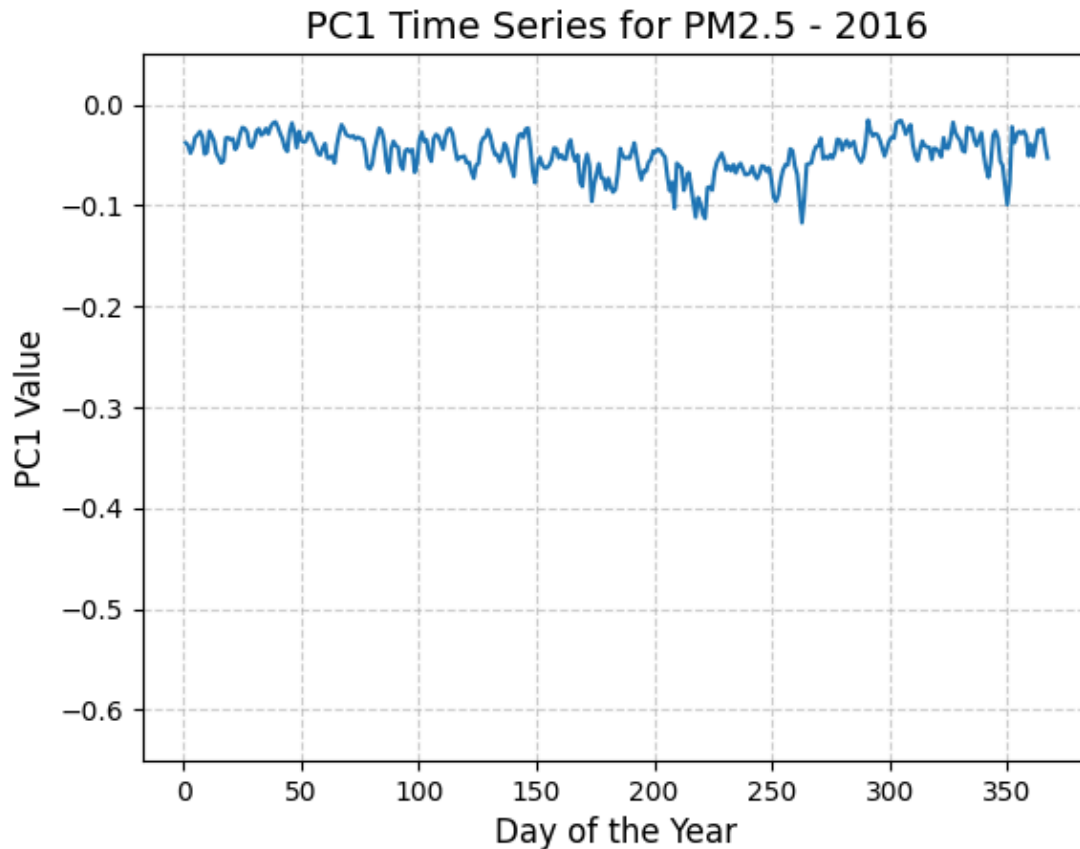


Figure 7: Temporal pattern for PM2.5 concentrations over the year of 2017.

It is important to note that the negative sign of our EOF and PC values is arbitrary and does not affect the interpretation of the patterns they represent. What matters is the relative magnitude and the consistency of signs across locations.

These results confirm that wildfire-affected counties were the primary contributors to elevated PM2.5 levels, particularly in the fire's active month.

CONCLUSION

Our analysis clearly demonstrates that the Thomas Fire significantly elevated PM_{2.5} concentrations in Santa Barbara and Ventura counties. The EOF results underscore the spatial dominance of these counties in shaping pollution patterns during the fire year.

While our study shows strong evidence of wildfire-linked PM_{2.5} increases by means of a robust methodology with spatial-temporal resolution, our data is limited to air quality and lacks health outcome data. Our data also gaps required imputation, which may introduce minor biases.

Future research should integrate public health data, particularly respiratory-related hospitalizations, to validate environmental-health correlations. Additionally, exploring meteorological data and smoke dispersion maps could enhance understanding of pollutant transport mechanisms.

DATA AND CODE AVAILABILITY

All PM_{2.5} concentration data were sourced from official California air quality databases. The

Python code used for analysis, data cleaning, and SVD computation is available at:

<https://github.com/ogenty001/Climate-Math-Research>

REFERENCES

- [1] : California Department of Forestry and Fire Protection (CAL FIRE). "Thomas Fire." CAL FIRE, 2017, <https://www.fire.ca.gov/incidents/2017/12/4/thomas-fire/> .
- [2] : Reid, Colleen E., et al. "Critical Review of Health Impacts of Wildfire Smoke Exposure." Environmental Health Perspectives, vol. 124, no. 9, 2016, pp. 1334–1343.
<https://doi.org/10.1289/ehp.1409277> .
- [3] : California Department of Forestry and Fire Protection. "2016 Fire Season Incident Archive." CAL FIRE, 2016, <https://www.fire.ca.gov/incidents/2016/> .
- [4] : U.S. Environmental Protection Agency (EPA). "Download Daily Data." EPA, <https://www.epa.gov/outdoor-air-quality-data/download-daily-data> .